

1) Store and Print Student Details - Basic Structure:

Input Format:

roll name age marks

Sample Input:

101 Arjun 19 87.5

Sample Output:

Roll No: 101

Name: Arjun

Age: 19

Marks: 87.50

PROGRAM:

```
#include <stdio.h>
```

```
struct student{
```

```
    int roll;
```

```
    char name[50];
```

```
    int age;
```

```
    int marks;
```

```
};
```

```
int main()
```

```
{
```

```
    struct student s1;
```

```
    printf("enter roll no:");
```

```
    scanf("%d",&s1.roll);
```

```
    printf("enter name:");
```

```
    scanf("%s",s1.name);
```

```
    printf("enter age:");
```

```
    scanf("%d",&s1.age);
```

```
    printf("enter marks:");
```

```
    scanf("%d",&s1.marks);
```

```
printf("roll:%d\n",s1.roll);
printf("name:%s\n",s1.name);
printf("age:%d\n",s1.age);
printf("marks:%d\n",s1.marks);

return 0;
}
```

2) Calculate Average Marks of Students - Array of Structures:

```
#include <stdio.h>

struct student{
    int roll;
    char name[50];
    int marks;
};

int main()
{
    int n;
    scanf("%d",&n);
    struct student s[n];
    for(int i=0;i<n;i++){
        printf("enter roll no:");
        scanf("%d",&s[i].roll);
        printf("enter name:");
        scanf("%s",s[i].name);
        printf("enter marks:");
        scanf("%d",&s[i].marks);
    }

    int sum=0;
```

```

for(int i=0;i<n;i++){
    sum+=s[i].marks;
}
float avg=sum/n;
printf("%.2f",avg);
return 0;
}

```

3) Student Address Details - Nested Structure:

```
#include <stdio.h>
```

```

struct student{
    int roll;
    char name[50];
    struct add{
        char city[50];
        int pin;
    }add;
};
int main()
{

```

```
    struct student s1;
```

```

    printf("enter roll no:");
    scanf("%d",&s1.roll);
    printf("enter name:");
    scanf("%s",s1.name);
    printf("enter address:");
    printf("enter city:");
    scanf("%s",s1.add.city);

```

```

printf("enter pin:");
scanf("%d",&s1.add.pin);

printf("roll no %d\n",s1.roll);
printf("name %s\n",s1.name);
printf("City %s\n",s1.add.city);
printf("Pincode %d",s1.add.pin);

return 0;
}

```

4) Update Student Marks - Pointer to Structure:

```

#include <stdio.h>
#include<stdlib.h>
struct student{
    int roll;
    char name[50];
    int mark;
    int bonus;
};
int main()
{

    struct student *s=(struct student*)malloc(sizeof(struct student));

    printf("enter roll no:");
    scanf("%d",&s->roll);
    printf("enter name:");
    scanf("%s",s->name);
    printf("enter mark:");
    scanf("%d",&s->mark);

```

```

    printf("enter bonus:");
    scanf("%d",&s->bonus);
    if((s->mark + s->bonus)>100){
        printf("100.00");
    }else{
        float ans=(s->mark + s->bonus);
        printf("%.2f",ans);
    }

    return 0;
}

```

5) Pass Structure to Function - Grade Finder:

```

#include <stdio.h>
#include<stdlib.h>
struct student{
    char name[50];
    int mark;
};
void ans(struct student *s){
    if(s->mark >=90){
        printf("A");
    }
    else if(s->mark >=75){
        printf("B");
    }
    else if(s->mark >=50){
        printf("C");
    }
    else{

```

```

        printf("F");
    }
}

int main()
{

    struct student *s=(struct student*)malloc(sizeof(struct student));

    printf("enter name:");
    scanf("%s",s->name);
    printf("enter mark:");
    scanf("%d",&s->mark);

    ans(s);
    return 0;
}

```

6) Return Structure from Function - Create Student Record:

```

#include <stdio.h>
#include<stdlib.h>
struct student{
    int roll;
    char name[50];
    float mark;
}s;

struct student ans(){
    printf("enter roll:");
    scanf("%d",&s.roll);
    printf("enter name:");
    scanf("%s",s.name);
}

```

```

    printf("enter mark:");
    scanf("%f",&s.mark);
    return s;
}
int main()
{
    ans();
    printf("Student created\n");
    printf("roll:%d\n",s.roll);
    printf("Name:%s\n",s.name);
    printf("Marks:%.2f",s.mark);

    return 0;
}

```

7) Student Percentage Using typedef Structure:

```

#include <stdio.h>
#include<stdlib.h>
typedef struct{
    int roll;
    char name[50];
    int mark1;
    int mark2;
    int mark3;
}student;
int main()
{

    student s;

```

```

printf("enter roll no:");
scanf("%d",&s.roll);
printf("enter name:");
scanf("%s",s.name);
printf("enter mark1:");
scanf("%d",&s.mark1);
printf("enter mark2:");
scanf("%d",&s.mark2);
printf("enter mark3:");
scanf("%d",&s.mark3);

int sum=s.mark1 + s.mark2 + s.mark3;
printf("Total:%d\n",sum);
float avg=sum/3;
printf("Percentage:%.2f",avg);

return 0;
}

```

8) Dynamic Student Records - malloc with Structure Array:

```

#include <stdio.h>
#include<stdlib.h>
struct student{
    int roll;
    char name[50];
    int marks;
};
int main()
{
    int n;
    scanf("%d",&n);

```



```

struct student *s=(struct student*)malloc(n * sizeof(struct student));
for(int i=0;i<n;i++){
    printf("student %d\n",i+1);
    printf("enter roll no:");
    scanf("%d",&s[i].roll);
    printf("enter name:");
    scanf("%s",s[i].name);
    printf("enter marks:");
    scanf("%d",&s[i].marks);
}

int sum=0;
for(int i=0;i<n;i++){
    sum+=s[i].marks;
}
float avg=sum/n;
printf("%.2f",avg);
return 0;
}

```

9) Student Fee Balance - Structure with Multiple Data Types:

```
#include <stdio.h>
```

```

struct student{
    int roll;
    char name[50];
    int tfee;
    int pfee;
};

int main()
{
    struct student s;

```

```

printf("enter roll no:");
scanf("%d",&s.roll);
printf("enter name:");
scanf("%s",s.name);
printf("enter total fees:");
scanf("%d",&s.tfee);
printf("enter paid fees:");
scanf("%d",&s.pfee);

float ans=s.tfee-s.pfee;
printf("%.2f",ans);
return 0;
}

```

10) Student Attendance Report - Array of Nested Structures:

```
#include <stdio.h>
```

```

struct student{
    int roll;
    char name[50];
    int tfee;
    int pfee;
};

int main()
{
    int n;
    scanf("%d",&n);
    struct student s[n];
    for(int i=0;i<n;i++){
        printf("enter roll no:");

```

```
scanf("%d",&s[i].roll);
printf("enter name:");
scanf("%s",s[i].name);
printf("enter total classes:");
scanf("%d",&s[i].tfec);
printf("enter attended classes:");
scanf("%d",&s[i].pfec);
}
for(int i=0;i<n;i++){
    float ans=((float)s[i].pfec/s[i].tfec)*100;
    printf("%s Attendance %.2f\n",s[i].name ,ans);
}
return 0;
}
```